

DRACARYS Platform - Development & Security Log

Date: September 25, 2026
Project: DRACARYS Web Platform
Role: AI Engineering Assistant & Lead Developer

Executive Summary

This log details the complete architectural overhaul, security hardening, and feature implementation of the DRACARYS platform. The goal was to transition the platform from a static, hardcoded application into a dynamic, production-grade enterprise system with robust admin controls and military-grade security.

1. Authentication & Core Architecture

- **Edge Middleware Overhaul:** Resolved an infinite login redirect loop by changing the Next-Auth strategy to JWT. This allowed Vercel's Edge runtime to read session cookies securely without needing direct database access.
- **Client-Side OAuth:** Replaced Server Action-based Google Sign-In with client-side signIn methods to permanently fix edge-network proxy hangs and timeouts.
- **Public Route Whitelisting:** Configured middleware to explicitly whitelist public marketing routes (/about, /team, /founder, /join, /hire, /contact) so visitors are never aggressively redirected to the login page.
- **Founder Override:** Engineered a permanent auth-bypass for the founder email (inayagamparthiban07@gmail.com). Upon login, this account automatically bypasses all queues and receives APPROVED and SUPER_ADMIN status.

2. Database Migration & Dynamic Team Architecture

- **Data Migration:** Executed a secure Node.js migration script to transfer 9 hardcoded demo members into the live PostgreSQL database.
- **Dynamic Frontend:** Completely rewrote the public /team page to fetch APPROVED members dynamically via Prisma. The team roster is now 100% database-driven.

3. Advanced Admin Controls ("God Mode")

- **Manage Members Portal:** Upgraded the "Users & Access" page into a comprehensive admin dashboard.
- **Instant Removal:** Added a secure Server Action to instantly delete users from the database, automatically removing them from the public frontend.
- **Edit Member Override:** Built a dedicated Edit Portal (/dashboard/admin/users/[id]/edit) allowing admins to manually override any user's Name, Member Tag, Tech Stack, Bio, Role, and Social Links.
- **Client-Side Form Resiliency:** Fixed "silent freezing" bugs on the Onboarding and Edit forms by converting them from raw HTML5 forms into interactive React Client Components with robust error handling and loading states.
- **Application Actions:** Replaced clunky HTML <select> menus with animated action buttons (Accept, Reject, Under Review) using React useTransition.

4. Automated Communications

- **Resend Integration:** Wired up the Admin Dashboard so that clicking "Accept" on a Join Application automatically sends a premium, HTML-formatted Welcome Email to the applicant, instructing them to log in.

5. Defense-in-Depth Security Hardening

- **Edge-Network Admin Protection:** Injected a security shield directly into middleware.ts. If a non-admin attempts to access /dashboard/admin/*, Vercel's global edge network instantly intercepts their token and rejects them before the server even processes the request.
- **Content Security Policy (CSP):** Implemented strict Next.js security headers to prevent Cross-Site Scripting (XSS) and unauthorized script execution.
- **Clickjacking Prevention:** Enforced X-Frame-Options: SAMEORIGIN and rame-ancestors 'none' to block malicious iframes.
- **IDOR Prevention:** Audited all Server Actions (like submitOnboarding) to ensure they strictly query against the authenticated session.user.id, mathematically preventing users from altering other people's profiles.

6. UI & UX Polish

- **Founder Page "Hero" Redesign:** Redesigned the /founder page into a premium "Hero Section" theme. Added glowing glass-morphism effects, pulsing online indicators, and centralized typography to match the homepage aesthetic.
- **Custom SVG Integrations:** Built and injected custom SVG icons (including a new Instagram icon) directly into the UI components for flawless scaling and performance.
- **Scrollbar Optimization:** Added custom global CSS to hide ugly default Windows scrollbars in the Admin Sidebar for a sleeker macOS-like feel.

Status: ALL SYSTEMS OPERATIONAL. CODEBASE IS PRODUCTION-READY.